Shanmugam Subramanian

Mechanical Engineer II | Sensor & RF Enclosure Design | Hyderabad, India

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SUMMARY

Mechanical engineer with 4 years of NPD and NPI experience designing enclosures, radomes and mounts for radar and RF sensing hardware. Developed IP68 (IEC 60529) radar chassis, cut internal temperature by 6 °C through fin optimization, and lowered radome cost by 8% without hurting dielectric performance. Experienced in tolerance stack-up, vibration and drop qualification, snap-fit design and FEA/CFD.

EXPERIENCE

JMA Wireless — Hyderabad, India
Mechanical Engineer II May 2023 – Present

- Led **electro-mechanical packaging** of eight 5G antenna products: released RF PCBs, sheet metal, die-cast, injection-molded, CNC, EDM, & FDM parts with a **90% FAI pass rate**.
- Performed **harmonic & random vibration analyses** with 80% first-pass correlation to physical tests (IEC 60068-2-6/64).
- Performed **IEC 60068-2-31 drop tests** for mechanical qualification of antenna products.
- Executed **static structural analysis** with FOS targets of 1.5–3 (metals) & 2–5 (plastics).
- Ran **non-linear (large-deformation) FEA** of a production snap-fit in Ansys Mechanical: predicted 38 N insertion and 30 N removal force and flagged a fillet redesign (FOS 1.13).
- Authored a **utility patent draft** for an injection-molded, snap-fit PCB/sheet-metal support pillar (twist-lock base, flexing offset arms, two-cavity mold).
- Designed pole, wall, ceiling, & tri-sector mounting brackets; gravity-assisted design **halved installation manpower**.
- Built Excel-based **hand-calculation tools** to estimate drag force from geometry, crimping force for pneumatic cylinder sizing, and snap-joint behavior in plastic part design.
- R&D of radome profiles using **Ansys Fluent CFD** (2D/3D); Design of experiments on 135+ profiles achieving **12–18% wind load reduction**. Documented simulation workflows for team adoption & working toward patent filing for further optimization.
- Developed a **pneumatic-actuated pivot punching fixture** that automated a 2-person task to one operator and **improved yield from 60% to 95%** (saving $0.4 per failed cycle).
- Automation & process-improvement by developing RF PCB sub-assembly fixtures, rivet press-pull tools, & alignment jigs, achieving **40% rework reduction, 75% faster riveting, and cutting up to 20–50 mins per antenna** in the production line.
- Managed **ECRs, FAIs, & SVPRs** using Salesforce Propel **PLM** & SolidWorks **PDM** in coordination with the global team.
- Supported **contract manufacturing** setup in Chennai through BOM costing, tooling transfer, & operator training.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Design Engineer (R&D) Jul 2022 – Apr 2023

- Developed two automotive radar antenna chassis and radomes with **IP68 (IEC 60529)** ratings for ADAS applications.
- Enhanced chassis thermal properties by refining fin geometry, reducing **internal temperature by 6°C**.
- Lowered radome **cost by 8%** through material optimization without degrading dielectric performance.
- Reverse-engineered & locally manufactured an RF waveguide connector, reducing cost by 8× compared to imports.

MulticoreWare India Pvt. Ltd. (Uhnder Team) — Chennai, India
Mechanical Project Intern Mar 2022 – Jun 2022

- Updated automotive radar housing designs based on lead engineer feedback & prepared GD&T-compliant drawings.
- Implemented **snap-fit** radome design, reducing the radar BOM by two components and saving $0.3 per unit.

EDUCATION

Bachelor of Engineering — Mechanical Engineering GPA: 8.21 / 10
Alagappa Chettiar College of Engineering and Technology (Anna University) 2018 – 2022

PROJECTS AND COMPETITIONS

SAE Bicycle Design Competition — 2020 - 2021 SAE INDIA
Developed a planetary gear-based bicycle drivetrain achieving 15% higher torque than a typical bicycle.
Mini Trencher Design — APRIL/ MAY - 2021 3rd Year Project B.E.
Designed a low-cost mini agricultural trencher; reduced 5× cost & size through drivetrain & structure optimization.
6-DOF Mimicking Robot— July 2025- Nov 2025 Personal project
Engineered a 6-DOF robotic arm with real-time mirroring, trajectory recording, and repeatable playback of arm movements.

SKILLS & SOFTWARE

Design: SolidWorks, AutoCAD, Fusion 360, Inventor, Onshape, **GD&T, Tolerance stackup**, enclosure & radome design, snap-fit design, 3D Printing
Simulation: Ansys Mechanical & Fluent, Finite Element Analysis, SolidWorks Simulation & Motion Analysis, thermal refinement
Programming & Embedded systems: Python, MATLAB Simulink & Simscape, C, C++, Arduino
Other Tools: Excel, PowerPoint, PLM (Salesforce Propel), SolidWorks PDM
Standards & Testing: IEC 60068-2-6/-64/-31 (vibration, drop), IEC 60529 (IP68), FAI, ECR